

The Deposits Channel of Monetary Policy

A Comprehensive Critical Review

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Abstract. This report provides a comprehensive critical review of Drechsler, Savov and Schnabl’s (2017) seminal paper proposing the *deposits channel* of monetary policy transmission. We examine the paper’s theoretical framework, identification strategy, and empirical findings in depth; evaluate its strengths, limitations, and key assumptions; situate it within the broader monetary economics and banking literature; and propose substantive extensions relevant to the post-fintech era. We further assess the channel’s contemporary relevance in light of the 2022–2023 Federal Reserve tightening cycle, the collapse of Silicon Valley Bank, and the prospective introduction of central bank digital currencies. We conclude that the deposits channel constitutes one of the most important empirical and theoretical contributions to monetary economics of the past two decades, while identifying meaningful limitations that future research must address.

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1. Introduction and Motivation

The question of how monetary policy transmits through the banking system to the real economy is one of the most consequential open problems in macroeconomics. The prevailing theoretical framework — the *bank lending channel* — held that the Federal Reserve controlled the aggregate supply of bank credit by regulating the quantity of required reserves. By expanding or contracting reserves, the Fed could directly influence the size of bank balance sheets and, through them, the volume of lending available to households and firms (Bernanke and Blinder, 1988).

This reserve mechanism, while theoretically clean, faced a fundamental empirical obstacle: required reserves had become quantitatively negligible relative to total bank assets by at least the 1980s. As documented by Romer and Romer (1990), Bernanke and Gertler (1995), and Woodford (2010), banks could easily circumvent reserve requirements by adjusting the composition of their liabilities. The problem became acute after 2008, when the Federal Reserve began paying interest on excess reserves, making reserve requirements entirely non-binding in equilibrium. The bank lending channel, in short, had compelling empirical support but no credible theoretical foundation for the modern institutional environment.

Drechsler, Savov and Schnabl (2017) — hereafter DSS — fill this gap with a new mechanism: the **deposits channel of monetary policy**. Their central claim is that banks possess significant *market power* over local deposit markets, and that this market power — rather than reserve requirements — is the mechanism through which changes in the Federal funds rate affect bank balance sheets and ultimately the real economy.

The paper's central research question can be stated as follows: *when the Fed funds rate rises, how do banks adjust the spreads they charge on deposits, how do depositors respond, and does the resulting funding contraction translate into a reduction in bank lending and economic activity?*

The intuition is elegant. When the Fed funds rate is low, holding cash is cheap, so households can easily substitute from deposits into cash. Banks must therefore keep deposit spreads low to avoid losing their deposit base. When the Fed funds rate rises, holding cash becomes expensive, which reduces the sensitivity of depositors to deposit rates — effectively granting banks greater market power. Banks exploit this by widening spreads, and depositors respond by gradually shifting into bonds and money market funds. Since deposits are a uniquely stable and low-cost source of bank funding that cannot be costlessly replaced by wholesale borrowing, the deposit contraction induces a contraction in lending. DSS estimate this channel is large enough to account for the *entire* monetary policy transmission through bank

balance sheets documented in the prior empirical literature.

This review proceeds as follows. Section 2 develops the economic intuition and theoretical framework. Section 3 describes the empirical strategy and identification approach. Section 4 summarises the main empirical findings. Section 5 provides a detailed critical evaluation of the paper's strengths and limitations. Section 6 situates DSS within the broader literature. Section 7 proposes substantive extensions. Section 8 evaluates the channel's contemporary relevance. Section 9 concludes.

2. Economic Intuition and Theoretical Framework

2.1 The Core Transmission Mechanism

The deposits channel operates through a sequence of interconnected steps that link the Fed funds rate to real lending. Let f denote the Fed funds rate and r_d the deposit rate. The *deposit spread* is defined as $s \equiv f - r_d$, representing the opportunity cost to depositors of holding deposits rather than bonds. The core empirical regularity motivating DSS's theory is that s increases substantially when f rises: for every 100 basis points (bps) increase in the Fed funds rate, the aggregate deposit spread widens by approximately 54 bps. This *incomplete pass-through* is the signature of bank market power in the deposit market.

The mechanism linking the Fed funds rate to deposit spreads operates through the outside option of depositors. Households derive liquidity from two main instruments: cash (currency and zero-interest accounts) and deposits. Cash provides maximal liquidity but pays no interest, so its opportunity cost equals the Fed funds rate f . Deposits are less liquid than pure cash but pay a positive rate $r_d = f - s$. Their opportunity cost relative to bonds is the spread s .

When f is low, cash is nearly costless to hold, making it a compelling outside option. Depositors can cheaply substitute from deposits into cash, so banks face highly elastic demand and must keep s low. When f is high, cash becomes expensive to hold (its opportunity cost is f), so the relevant outside option for depositors shifts from cash to bonds. Since bonds provide no liquidity services and banks face limited within-market competition in concentrated markets, deposit demand becomes more inelastic. Banks can therefore widen s without incurring proportionate outflows. The key insight is that **bank market power is endogenous to the level of the Fed funds rate**.

The Deposits Channel: Step-by-Step

1. $f \uparrow \Rightarrow$ cost of holding cash rises
2. Depositors less likely to substitute from deposits into cash
3. Deposit demand becomes more inelastic; banks face less competition from cash
4. Banks widen deposit spreads s (keep r_d low relative to f)
5. Even though inelastic, depositors gradually shift to bonds/money markets
6. Aggregate deposit outflows reduce banks' stable funding base
7. Deposits cannot be fully replaced by costly wholesale funding
8. Banks contract lending $\Rightarrow$ real economic activity falls

2.2 The Theoretical Model*2.2.1 Household Preferences*

The representative household maximises utility over final wealth W and liquidity services l according to a constant elasticity of substitution (CES) aggregator:

$$u(W_0) = \max \left[W^{\frac{\rho-1}{\rho}} + \lambda l^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}}, \quad (1)$$

where $\lambda > 0$ is a share parameter and $\rho < 1$ ensures that wealth and liquidity are complements — households value liquidity services *in addition to* financial returns, not as a substitute for them. This preference structure is micro-founded either by a cash-in-advance constraint (Gali, 2009) or by a preference for extreme safety (Stein, 2012).

Liquidity services are themselves derived from cash M and deposits D via a second CES aggregator:

$$l(M, D) = \left[M^{-\frac{1}{\epsilon}} + \delta D^{-\frac{1}{\epsilon}} \right]^{-\epsilon}, \quad (2)$$

where $\epsilon > 1$ ensures cash and deposits are substitutes in the provision of liquidity, and $\delta \in (0, 1)$ captures the fact that deposits are less liquid than cash. Deposits at different banks are themselves imperfect substitutes, with elasticity of substitution $\eta > 1$ across institutions, which gives individual banks market power.

2.2.2 Equilibrium Deposit Spreads

Each bank i sets its deposit spread s_i to maximise deposit profits $D_i s_i$, subject to the household's demand system. In a symmetric equilibrium with N banks, the profit-maximising condition reduces to equating the elasticity of demand for bank

i 's deposits to -1 . Solving this optimality condition yields the equilibrium deposit spread:

$$s = \delta^{\frac{1}{\epsilon-1}} \cdot \frac{\mathcal{M} - \rho}{\epsilon - \mathcal{M}} \cdot f, \quad (3)$$

where the market power parameter is:

$$\mathcal{M} \equiv 1 - (\eta - 1)(N - 1). \quad (4)$$

Equation (4) shows that market power $\mathcal{M}$ is higher when there are fewer banks (N small) or when depositors substitute less readily across banks (η close to 1). In the limiting case of a monopolist, $\mathcal{M} = 1$.

From equation (3), three key theoretical predictions follow directly:

1. The deposit spread s is *proportional* to the Fed funds rate f , with factor:

$$\frac{\mathcal{M} - \rho}{\epsilon - \mathcal{M}} \delta^{1/(\epsilon-1)}.$$

2. The *deposit spread beta*, defined as $\beta^s \equiv \partial s / \partial f$, is strictly increasing in market power $\mathcal{M}$.
3. When f is high, the aggregate elasticity of deposit demand approaches ρ (the low wealth–liquidity substitution elasticity) because cash is expensive; when f is low, it approaches ϵ (the higher cash–deposit substitution elasticity). Thus market power is endogenous to the rate level.

2.2.3 Financial Sophistication as an Additional Source of Market Power

Geographic concentration is only one source of bank market power. DSS extend the model (Proposition 2) to allow for a fraction α_{ns} of “nonswitchers” — depositors who do not monitor rates at competing institutions and therefore do not respond to spread differentials. The equilibrium deposit spread beta is preserved in form, but with market power parameter generalised to:

$$\mathcal{M}_{\text{ns}} = 1 - (\eta - 1) \left[\frac{1}{\alpha_{\text{ns}} + (1 - \alpha_{\text{ns}}) \frac{1}{N}} - 1 \right]^{-1}. \quad (5)$$

This extension yields several important implications. First, the deposit spread beta increases with the proportion of nonswitchers, as they represent an additional source of insulated market power. Second, even in markets with zero concentration (as $N \rightarrow \infty$), a sufficiently large fraction of nonswitchers can sustain positive and economically meaningful spread betas. Third, and most importantly, this result establishes that the deposit spread beta is a *sufficient statistic* for aggregate market power — it captures the combined effect of concentration, switching costs, financial

inattention, and depositor sophistication in a single observable metric.

2.2.4 Effects on Lending and Wholesale Funding

DSS extend the baseline model (Proposition 3) to incorporate profitable lending opportunities (with decreasing marginal returns at rate l_1) and costly wholesale (non-deposit) funding (with cost increasing at rate h). Each bank i now solves:

$$\max_{D_i, H_i} \left[f + l_0 - \frac{l_1}{2} L_i \right] L_i - \left[f + \frac{h}{2} H_i \right] H_i - (f - s_i) D_i, \quad (6)$$

where $L_i = H_i + D_i$ is total lending, H_i is wholesale funding, and $l_0, l_1, h > 0$.

The first-order condition for wholesale funding implies:

$$H_i = \frac{l_0}{l_1 + h} - \frac{l_1}{l_1 + h} D_i. \quad (7)$$

Equation (7) shows that banks partially offset a decline in deposits by increasing wholesale funding. However, since wholesale funding is costly ($h > 0$), this offset is only partial — the Modigliani–Miller theorem fails in this economy. Total lending is therefore:

$$L_i = \frac{l_0}{l_1 + h} + \frac{h}{l_1 + h} D_i, \quad (8)$$

which is increasing in deposit volume. A contraction in deposits therefore induces a contraction in lending, with magnitude governed by the ratio $h/(l_1 + h)$. When h is large (wholesale funding is very costly), the pass-through from deposits to lending is near-complete; when h is small, banks can easily substitute and lending is relatively insulated.

Proposition 3 of DSS establishes that in this enriched model, all key predictions from the baseline are preserved: higher rates increase the effective market power of banks (raising s), inducing deposit contraction, partial wholesale funding substitution, and a net reduction in lending.

3. Empirical Strategy and Identification

3.1 Data Sources

DSS construct their empirical analysis from five primary datasets covering the United States from 1994 to 2013:

- **FDIC Summary of Deposits:** Annual branch-level deposit quantities for the universe of U.S. bank branches (June 1994–June 2014). Provides geographic

coordinates, parent bank identifiers, and total deposit volumes.

- **Ratewatch:** Weekly branch-level deposit rates by product type, covering January 1997 to December 2013 and approximately 54% of all U.S. bank branches. Focused analysis is on the two most widely-offered products: \$25,000 money market deposit accounts (savings proxy) and \$10,000 12-month certificates of deposit (time deposit proxy).
- **U.S. Call Reports:** Quarterly bank-level income statements and balance sheets for all U.S. commercial banks (Federal Reserve Bank of Chicago, January 1994–December 2013).
- **NCRC Small Business Lending:** Annual new small business loan originations (loans < \$1 million) at the bank-county level from 1997 to 2013.
- **FRED and Datastream:** Fed funds target rate, one-year T-bill rate, and Fed funds futures prices for expected-rate decomposition.

Market concentration is measured using the Herfindahl-Hirschman Index (HHI), defined for county c in year t as:

$$\text{HHI}_{ct} = \sum_{j=1}^{N_{ct}} \left(\frac{D_{jct}}{D_{ct}} \right)^2, \quad (9)$$

where D_{jct} is bank j 's deposits in county c at time t , and D_{ct} is total county deposits. Branch-HHI is the time-averaged value of equation (9) for the county where a branch is located. Bank-HHI is the deposit-weighted average of Branch-HHI across all of a bank's branches.

The resulting cross-sectional variation is substantial. Branch-HHI ranges from 0.06 in the most competitive urban markets to 1.0 in monopolistic rural counties, with a mean of 0.35 and standard deviation of 0.20 across counties.

3.2 The Identification Challenge

The central identification problem is that deposit supply and banks' lending opportunities may be simultaneously affected by macroeconomic conditions. When the Fed raises rates in response to an economic expansion, loan demand may also be elevated, causing banks to take in more deposits independently of any deposit channel. Conversely, if tightening accompanies a downturn, declining loan demand may reduce deposit-taking even in the absence of market power effects. A simple time-series regression of deposit growth on the Fed funds rate cannot disentangle these supply and demand effects.

3.3 Within-Bank, Across-Branch Estimation

DSS's primary solution is a within-bank, across-branch identification strategy that exploits geographic variation in deposit market concentration. The fundamental observation is that since banks operate internal capital markets — raising deposits at some branches and lending them at others — the deposit decision at a given branch is separable from local lending decisions. By comparing branches of the *same bank* located in counties with *different* HHI levels, DSS can control for the bank's lending opportunities while estimating the pure market power effect.

The baseline panel regression is:

$$\Delta y_{it} = \alpha_i + \zeta_{c(i)} + \lambda_{s(i)t} + \delta_{j(i)t} + \gamma \cdot \Delta FF_t \times \text{Branch-HHI}_i + \varepsilon_{it}, \quad (10)$$

where:

- Δy_{it} is either the change in the deposit spread or the log change in total deposits at branch i from t to $t + 1$;
- ΔFF_t is the contemporaneous change in the Fed funds target rate;
- Branch-HHI_i is the time-averaged HHI of the county where branch i is located;
- $\delta_{j(i)t}$ are **bank $\times$ time fixed effects** for the bank j that owns branch i — the key control that absorbs all time-varying bank-specific factors, including lending opportunities;
- α_i , $\zeta_{c(i)}$, and $\lambda_{s(i)t}$ are branch, county, and state $\times$ time fixed effects, respectively;
- Standard errors are clustered at the county level.

The coefficient of interest is γ : the differential sensitivity of deposit spreads (or flows) to monetary policy, as a function of local market concentration. The identifying assumption is that banks equalise marginal returns to lending across their branches, so that the deposit allocation decision at each branch is independent of local lending conditions. This assumption is supported by the within-county lending results in Section 7 of DSS, which show that a bank's lending in a given county is unrelated to the local concentration of deposit markets (supporting internal fund allocation), and by external evidence from [Gilje, Loutskina and Strahan \(2016\)](#) showing that banks channel deposits toward high-demand lending markets.

3.4 Event Study: Timing of Deposit Spread Adjustment

To provide sharper causal identification, DSS exploit the weekly frequency of Rate-watch data to conduct an event study around FOMC announcements. The regression is:

$$\Delta y_{it} = \alpha_t + \zeta_{c(i)} + \sum_{\tau=-5}^5 \gamma_{\tau} \cdot \text{Branch-HHI}_{c(i)} \times \Delta FF_{t-\tau} + \varepsilon_{it}, \quad (11)$$

where the five leads and lags of ΔFF encompass the full six-week FOMC decision cycle. The cumulative sum $\sum_{\tau=-5}^t \gamma_{\tau}$ traces the differential spread response at high- versus low-concentration branches around week 0 (the announcement week).

If the deposits channel is the true mechanism, we expect: (i) no differential pre-announcement drift (ruling out anticipation effects); (ii) a sharp response beginning in week 0 (consistent with a direct monetary policy effect); and (iii) a permanent level shift thereafter (consistent with the quarterly regression magnitudes).

3.5 Decomposition into Expected and Unexpected Rate Changes

A further identification test exploits the fact that savings deposits have zero maturity, so their rates should respond to a rate change *when it occurs* regardless of whether it was anticipated. This contrasts with long-lived assets like stocks and bonds, which incorporate expected rate changes in advance and react only to surprises. DSS replace the total rate change in equation (10) with its expected component (measured from Fed funds futures prices three months prior) and the unexpected residual, finding that both components have quantitatively similar effects on deposit spreads. This decomposition rules out the interpretation that the results are driven by information released at the time of FOMC decisions rather than by the rate change itself.

3.6 Lending Channel Identification: Within-County Estimation

For the lending analysis, banks can allocate deposits across branches, so the within-bank identification strategy used for deposits is inapplicable. Instead, DSS compare the lending behaviour of different banks within the same county (same lending market) while varying their bank-level deposit concentration (Bank-HHI). The lending regression is:

$$\Delta \log(\text{lending})_{jct} = \alpha_{jc} + \delta_{ct} + \beta \cdot \text{Bank-HHI}_{jt-1} + \gamma \cdot \Delta FF_t \times \text{Bank-HHI}_{jt-1} + \varepsilon_{jct}, \quad (12)$$

where α_{jc} are bank $\times$ county fixed effects and δ_{ct} are county $\times$ time fixed effects that absorb all local lending demand changes. The coefficient γ identifies whether, controlling for local demand, banks with more concentrated deposit bases (higher Bank-HHI) reduce lending more when the Fed funds rate rises.

4. Key Empirical Findings

4.1 Aggregate Time-Series Evidence

The aggregate time series provides compelling prima facie evidence for the deposits channel. Figure I of DSS plots the average deposit rate against the Fed funds target rate from 1986 to 2013, revealing a striking pattern of incomplete pass-through: deposit rates consistently track the Fed funds rate, but with a systematic lag and shortfall.

Aggregate Spread Evidence

For every 100 bps increase in the Fed funds rate, the aggregate deposit spread widens by **54 bps**. During the 425 bps mid-2000s hiking cycle, the deposit spread rose by 245 bps. Spreads on savings and checking deposits increased by 340 bps and 470 bps respectively, while time deposit spreads rose by only 105 bps — consistent with greater market power over more liquid deposit categories.

Figure II of DSS documents deposit quantity responses. Core deposit growth is strongly negatively correlated with changes in the Fed funds rate (correlation = -49%), with year-over-year growth rates ranging from -1% to $+18\%$ over the sample. The correlations are stronger for liquid categories (savings: -59% , checking: -33%) and positive for illiquid time deposits ($+23\%$), reflecting within-deposit substitution as liquid instruments become relatively more expensive.

Crucially, the simultaneous observation of rising spreads and falling deposit quantities identifies a *supply curve shift*: if the effect were driven by a demand shift, prices and quantities would move in the same direction.

4.2 Branch-Level Within-Bank Results

Table 1 summarises the within-bank estimation results from equations (10) for deposit spreads and flows.

Table 1: Deposit Spreads, Flows, and Monetary Policy: Within-Bank Estimates

Specification	Spread Response		Flow Response
	Savings Δ Spread	Time Δ Spread	$\Delta \log(\text{Deposits})$
$\Delta FF \times \text{Branch-HHI}$	0.140*** (0.033)	0.074*** (0.025)	-0.661*** (0.254)
Bank $\times$ Quarter FE	✓	✓	✓
State $\times$ Quarter FE	✓	✓	✓
Branch FE	✓	✓	✓
County FE	✓	✓	✓
Observations	117,683	121,990	1,150,049
R^2	0.810	0.808	0.344

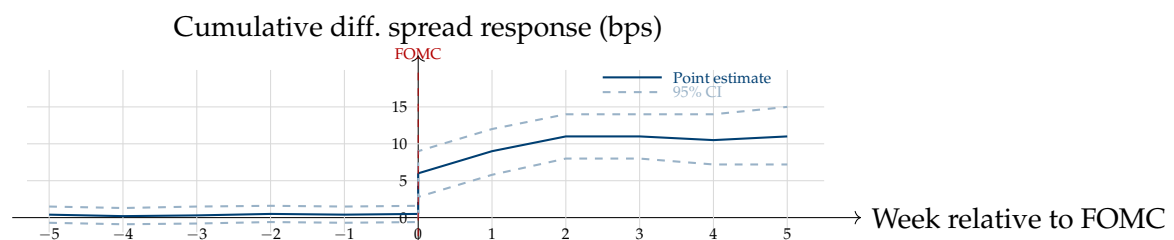
Notes. Standard errors clustered at county level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Preferred specification with full set of fixed effects shown. Source: DSS (2017), Tables II and III.

Following a 100 bps increase in the Fed funds rate, branches of the same bank in high-concentration counties raise savings deposit spreads by **14 bps more** and experience deposit outflows **66 bps larger** than branches in low-concentration counties. The results are robust across all specification variants, including with and without state $\times$ time fixed effects.

The implied semi-elasticity of deposits with respect to deposit spreads is -5.3 , meaning a 100 bps increase in the deposit spread is associated with a 530 bps contraction in deposits. Since a 100 bps rate hike generates on average a 61 bps spread increase among large banks, the implied deposit contraction per 100 bps hike is approximately 323 bps.

4.3 Event Study Results

Figure V of DSS (reproduced schematically as Figure 1 below) provides some of the most compelling evidence in the paper.

**Figure 1:** Event-study differential deposit spread response (DSS, Figure V).

The event study confirms: (i) zero differential pre-announcement response, ruling

out anticipation and slow-moving confounds; (ii) a jump of approximately 6 bps in the week of the FOMC decision; (iii) accumulation to ≈ 11 bps over the following two weeks; and (iv) a permanent level shift consistent with the quarterly regression estimates.

4.4 Financial Sophistication

Table V of DSS confirms that financial sophistication amplifies market power independently of concentration. Using county-level proxies — share of population aged 65+ (age), log median household income (income), and share with a college degree (education) — DSS find that branches in counties with older, lower-income, or less-educated populations raise spreads more and experience greater outflows when rates rise, all else equal. College education is the most robust proxy when all three are included jointly. These findings validate Proposition 2: the deposit spread beta is a sufficient statistic for market power from both structural (concentration) and behavioral (sophistication) sources.

4.5 Lending and Real Economic Activity

Table 2: Deposits Channel Effects on Lending and Real Activity

Outcome	Coefficient	Significance
<i>Bank-county lending (within-county estimation)</i>		
$\Delta FF \times \text{Bank-HHI}$ (new SB lending)	−0.208 per unit HHI	**
Per 1 SD increase in Bank-HHI	−291 bps per 100 bps hike	**
<i>County-level outcomes (within-county estimation)</i>		
$\Delta FF \times \text{County-HHI}$ (new lending)	−0.093	***
Per 1 SD increase (58 bps per 100 bps)		
$\Delta FF \times \text{County-HHI}$ (employment Δ)	−0.014	***
Per 1 SD increase (−9 bps per 100 bps)		
$\Delta FF \times \text{County-HHI}$ (wage bill Δ)	−0.011	***
Per 1 SD increase (−11 bps per 100 bps)		

Notes. *** $p < 0.01$, ** $p < 0.05$. SB = small business. Standard errors double-clustered at bank and county level (lending regressions) or clustered at county level (county-level regressions). Source: DSS (2017), Tables VI and VII.

Table 2 shows that the deposit contraction has economically and statistically significant effects on lending and employment. Small business lending is particularly sensitive because it is illiquid and risky, making it highly dependent on the stable funding that deposits uniquely provide.

The bank-level Call Reports analysis (Table VIII of DSS) corroborates these findings. Banks with higher Bank-HHI experience greater deposit outflows, partially offset by increased wholesale funding, with a net contraction in total liabilities. Assets decline correspondingly, driven by reductions in securities, real estate loans, and commercial and industrial loans.

4.6 Aggregate Scale and the Deposit Spread Beta

The deposit spread beta, estimated by regressing each bank's deposit spread on the Fed funds rate (with up to four lags to allow for adjustment in non-zero-maturity products), averages 0.54 across all banks and 0.61 for the largest 5% of banks. The cross-sectional distribution is wide, ranging from 0.31 at the 10th percentile to 0.89 at the 90th percentile.

Aggregate Quantitative Implications

Outcome (400 bps hiking cycle)	Percentage	Dollar value (2014 base)
Deposit reduction	−1,458 bps	\$1.35 trillion
Lending reduction	−995 bps	\$763 billion

Comparison to Bernanke & Blinder (1992) VAR (per 31 bps hike):

Deposit outflows: B&B = 81 bps; DSS = 113 bps

Securities reduction: B&B = 123 bps; DSS = 154 bps

Loan contraction: B&B = 57 bps; DSS = 77 bps

DSS estimates are large enough to account for the *entire* transmission of monetary policy through bank balance sheets documented in the prior literature — without requiring any role for reserve requirements.

4.7 The Liquidity Premium

Since deposits are the primary source of liquid assets for households, their contraction mechanically reduces the aggregate supply of liquidity in the economy. DSS document a striking 90% correlation between the aggregate deposit spread and the T-bill liquidity premium (defined as the Fed funds rate minus the three-month T-bill rate) over 1986–2013. This relationship is strong both at business cycle frequencies and in long-run trends.

This finding connects the deposits channel to the broader asset pricing literature. When monetary policy tightens and deposit spreads widen, the supply of house-

hold liquidity contracts, pushing up the liquidity premium on all safe and liquid assets including Treasury securities. This provides a supply-side explanation for the puzzling tight relationship between the Fed funds rate and the liquidity premium documented by Nagel (2016), which had previously lacked a micro-founded theoretical account.

5. Critical Evaluation

5.1 Strengths

5.1.1 Identification Strategy

The within-bank, across-branch design is among the most credible natural experiments in the empirical banking literature. By holding the bank fixed (through bank $\times$ time fixed effects) and exploiting plausibly exogenous geographic variation in market concentration, DSS cleanly isolate the market power mechanism from the confounding effect of lending opportunities. The event study's week-zero timing response closes off virtually all remaining alternative explanations. These two identification strategies are mutually reinforcing rather than redundant: the quarterly panel regression establishes the magnitude and persistence of the concentration effect, while the event study pins down precise causal timing.

5.1.2 Empirical Breadth and Internal Consistency

Results are consistent across five levels of analysis — aggregate time series, branch-level spread and flow regressions, bank-level Call Reports, county-level employment outcomes, and the event study around FOMC announcements — using three independent datasets (Ratewatch, FDIC, Call Reports). Quantitative magnitudes are also internally consistent: the within-bank semi-elasticity (-5.3) and the spread-beta-based estimate (-3.6 for all banks, -6.0 for large banks) bracket a plausible range, and both align with the aggregate time-series patterns. This multi-level consistency is a powerful indicator of a genuine mechanism rather than a statistical artefact.

5.1.3 The Deposit Spread Beta as a Policy Instrument

The deposit spread beta is a theoretically grounded, easily observable, bank-level measure of monetary policy exposure that aggregates all sources of market power — concentration, switching costs, financial sophistication, product differentiation — into a single sufficient statistic. This has genuine practical value for bank supervisors and macroprudential regulators seeking to identify which institutions are most

exposed to rate cycles. It also provides a natural metric for stress testing: a bank with a high spread beta will experience larger deposit outflows and greater balance sheet contraction for a given tightening episode.

5.1.4 Resolution of a Long-Standing Theoretical Puzzle

By providing a market-power foundation for the bank lending channel, DSS resolve a problem that had persisted in monetary economics for over three decades. The mechanism requires no reserves, no binding capital constraints, and no assumption about long-term rate movements — it operates through the level of the short rate, which is precisely the instrument the Fed controls. This also provides theoretical grounding for the observed practice of gradual rate adjustment, since each rate change has a direct and meaningful effect on deposit supply independent of its level relative to the natural rate.

5.2 Weaknesses and Limitations

Representative Household Assumption

The CES framework aggregates all depositors into a single representative agent with constant elasticity preferences. This fundamentally cannot capture the heterogeneity in depositor behaviour that drives the tail risks most relevant to financial stability. Wealthy, financially sophisticated depositors — corporate treasurers, institutional cash managers, and high-net-worth individuals — have near-zero switching costs and near-infinite elasticity. Retail depositors exhibit high inertia, inattention, and genuine switching costs. Aggregating these populations produces average parameters that may accurately predict gradual, economy-wide adjustment while completely missing the concentrated, rapid outflows that precipitate bank runs. The Silicon Valley Bank episode (Section 8) illustrates precisely this limitation.

Pre-Fintech Sample Period: 1994–2013

The empirical analysis predates the structural disruption of digital banking. The rise of online high-yield savings accounts (e.g., Marcus by Goldman Sachs, Ally Bank), mobile banking applications, and fintech platforms from approximately 2015 onward has dramatically reduced geographic switching frictions. A depositor in 2024 can open a new account in minutes on a smartphone, rendering county-level HHI a less meaningful proxy for competitive pressure than it was in 2005. The model's market power parameters — and therefore its quantitative predictions for the magnitude and speed of the deposits channel — likely

require substantial re-estimation for the post-2015 era. The 2022–2023 hiking cycle, discussed in Section 8, provides preliminary evidence of faster adjustment consistent with reduced switching costs.

Wholesale Funding Cost: Assumed Rather Than Estimated

The parameter h governing the marginal cost of wholesale funding plays a central role in translating deposit outflows into lending contraction (equation 8), but is not directly estimated. In practice, this cost varies substantially with macroeconomic conditions, a bank's credit rating, the regulatory environment (Basel III's Liquidity Coverage Ratio and Net Stable Funding Ratio impose explicit costs on wholesale reliance), and prevailing stress in money markets. Treating h as a fixed structural constant limits the model's portability across different regulatory regimes and time periods, and introduces unquantified sensitivity in the lending contraction estimates.

Static and Geographic Concentration Measure

Branch-HHI is computed as a time-averaged, county-level measure that abstracts from dynamic competitive entry, the growing role of non-bank competitors (money market funds, brokerage platforms, fintech lenders), and cross-geographic competition from internet banks. In the modern era, a bank's true competitive environment in deposit-taking is increasingly defined by product characteristics and digital accessibility rather than geographic proximity. The assumption that county-level HHI adequately proxies for a bank's market power becomes increasingly tenuous as the share of online deposit-taking grows.

General Equilibrium and Endogeneity of Aggregate Estimates

DSS acknowledge that because the Fed tends to raise rates during economic expansions, aggregate lending at the time of a hiking cycle is elevated by high loan demand, partially offsetting the deposit-channel-induced contraction. The cross-sectional identification controls for this at the bank and county level, but the aggregate dollar estimates (\$1.35T deposits, \$763B lending) are necessarily computed under the counterfactual assumption that lending would have remained unchanged. These are therefore best interpreted as upper bounds on the net contractionary effect of a tightening cycle.

5.3 Assessment of Key Assumptions

Table 3: Assessment of Core Modelling and Identification Assumptions

Assumption	Role in Paper	Assessment
Banks freely allocate deposits across branches	Enables within-bank identification	Indirectly validated by lending regressions; may fail for small single-market banks (~30% of sample)
Wealth and liquidity are complements ($\rho < 1$)	Drives demand in-elasticity at high rates	Well-grounded in cash-in-advance and safety-preference frameworks; validated by liquid vs. illiquid deposit differential
Cash and deposits are substitutes ($\epsilon > 1$)	Ensures households switch to cash when rates are low	Consistent with near-zero cash demand at low rates in equilibrium
Market power stable over time	Justifies time-averaged HHI measure	Increasingly tenuous post-2015; digital banking disrupts geographic concentration
Wholesale funding cost h is constant	Governs lending channel magnitude	Varies with regulation, stress, and monetary regime; not estimated
Representative household	Enables clean aggregation	Problematic for heterogeneous depositor dynamics and tail risk (SVB, bank runs)

6. Engagement with the Broader Literature

6.1 The Bank Lending Channel

The deposits channel directly addresses the long-standing theoretical gap in the bank lending channel literature initiated by [Bernanke and Blinder \(1988\)](#) and elaborated by [Kashyap and Stein \(1994\)](#), [Kashyap, Stein and Wilcox \(1993\)](#), and [Kashyap and Stein \(2000\)](#). These papers documented robust empirical relationships between monetary policy and bank credit supply but relied on reserve requirements as the transmission mechanism — a foundation that had become implausible by the 1990s. DSS provide the new micro-foundation the literature needed. Quantitatively, their estimates bracket the VAR-based estimates of [Bernanke and Blinder \(1992\)](#), with

their aggregate figures accounting for the full magnitude of bank balance sheet transmission.

6.2 The Balance Sheet Channel

The deposits channel makes a sharply different prediction from the balance sheet or net worth channel of [Bernanke and Gertler \(1989\)](#), [Kiyotaki and Moore \(1997\)](#), and [Brunnermeier and Sannikov \(2014\)](#). Under balance sheet amplification, a surprise increase in long-term interest rates reduces bank net worth, forcing a symmetric contraction in all liabilities. Under the deposits channel, banks specifically *reduce deposit funding and increase wholesale funding* — a substitution that is consistent with what DSS observe empirically. This differential prediction provides clean discriminating power between the two channels.

6.3 The New Keynesian Framework

DSS complement rather than replace the New Keynesian model ([Woodford, 2003](#)), but with two important distinctions. In the standard New Keynesian framework, the short rate matters only insofar as it influences long-term rates through the expectations hypothesis. In the deposits channel, the short rate matters *in its own right* because it directly affects the supply of liquid assets and the cost of bank funding. This provides a theoretical basis for the empirical observation that the level of the short rate — not just its deviation from the neutral rate — influences economic outcomes, and for the practice of gradual rate adjustment ([Bernanke, 2004](#); [Stein and Sunderam, 2015](#)).

6.4 Deposit Pricing and Market Structure

The deposit pricing literature ([Hannan and Berger, 1989, 1991](#); [Neumark and Sharpe, 1992](#); [Driscoll and Judson, 2013](#)) documents that deposit rate adjustment is slow and asymmetric, particularly in concentrated markets, and interprets this as evidence of New Keynesian price rigidities. DSS depart from this interpretation in two important ways: they focus on permanent changes in the *level* of deposit spreads (not the adjustment speed), and they jointly analyse deposit quantities, which prior work largely ignored. This joint analysis of prices and quantities is essential for identifying supply versus demand effects and for quantifying the aggregate implications of the channel.

6.5 The Liquidity Premium Literature

[Nagel \(2016\)](#) documents a puzzling 90% correlation between the Fed funds rate and the T-bill liquidity premium over several decades, but does not provide a micro-

founded supply-side explanation. DSS fill this gap: by showing that deposit supply contracts when rates rise, they explain how monetary policy mechanically reduces aggregate household liquidity and pushes up the liquidity premium on all safe assets. This connects monetary economics to the asset pricing literature on the convenience yield of near-money assets (Krishnamurthy and Vissing-Jørgensen, 2012), suggesting that monetary transmission operates not just through credit supply but through liquidity pricing across all financial markets.

7. Extensions and Avenues for Future Research

Extension 1: Digital Banking and the Structural Erosion of Market Power

Hypothesis. The rise of online high-yield savings platforms and mobile banking from ~2015 onward has structurally reduced geographic switching frictions, compressing deposit spread betas toward zero.

Empirical design. Exploit the staggered geographic rollout of major online savings products (e.g., Marcus by Goldman Sachs, launched 2016; Ally Bank's mobile app penetration) as a quasi-exogenous shock to the competitive environment faced by incumbent brick-and-mortar banks. Estimate whether spread betas and the sensitivity of deposit outflows to rate changes declined following exposure to online bank entry. Use county-level smartphone adoption rates and broadband penetration as instrumental variables for the timing of digital banking disruption.

Expected finding. Spread betas should have declined monotonically with digital bank market share, particularly in previously high-HHI counties where the disruption was most pronounced. If confirmed, this implies that the deposits channel has been weakening as a monetary transmission mechanism precisely as the Fed has become more reliant on rate-based policy — with potential implications for the size and duration of rate cycles required to achieve a given macroeconomic effect.

Extension 2: Central Bank Digital Currency and the Deposit Channel

A retail CBDC paying a rate mechanically linked to monetary policy would constitute direct government competition with bank deposits. This creates fundamentally opposing forces for the deposits channel.

Amplification effect. With a CBDC available at zero switching cost and government backing, depositors would immediately and fully substitute out of bank deposits when rates rise and the CBDC rate adjusts instantaneously. Pass-through

would approach unity, accelerating and intensifying the deposit channel's operation.

Destruction effect. The existence of the CBDC as a permanent outside option would eliminate banks' market power in deposit markets entirely — driving spread betas toward zero in steady state, collapsing deposit franchise values, and potentially triggering systematic bank disintermediation.

Model extension. Introduce the CBDC as a fourth asset class in the household's liquidity aggregator (equation 2), paying rate $r_{\text{CBDC}} = f - s_{\text{CBDC}}$ where s_{CBDC} is a small fixed spread set by the central bank. Solve for equilibrium bank spreads and deposit quantities as a function of CBDC adoption rates. Quantify the tradeoff between faster policy transmission and the financial stability risk of rapid bank disintermediation during tightening cycles.

Extension 3: Heterogeneous Depositor Model

Replace the representative household in equations (1)–(2) with a distribution of depositor types indexed by sophistication parameter $\theta_i \sim F(\theta)$, where θ_i governs both switching costs and awareness of competing rates.

Key predictions of the extended model.

- Aggregate deposit elasticity becomes a non-linear function of the Fed funds rate, depending on which segment of the θ -distribution is marginal at any given rate level.
- Bank run dynamics emerge naturally: when a sufficiently negative information shock causes sophisticated depositors (high θ_i) to withdraw en masse, uninformed depositors update their beliefs and run dynamics can occur even without insolvency.
- The model generates distributional predictions: monetary tightening disproportionately benefits banks serving unsophisticated depositor bases, which captures more rents; these same banks are most exposed to run risk if depositor composition shifts.

Estimation. Linked administrative bank account and household wealth data — available in Europe (ECB's Household Finance and Consumption Survey combined with supervisory data) and Canada (Bank of Canada administrative records) — would enable structural estimation of the θ -distribution and a proper test of the heterogeneous extension.

Extension 4: Asymmetric Pass-Through Across the Rate Cycle

DSS document the deposits channel primarily during rate-hiking episodes. A natural question is whether the channel operates symmetrically during easing cycles. Existing evidence on deposit pricing rigidity (e.g., [Hannan and Berger 1991](#)) suggests that rates are downwardly rigid: banks are slow to raise deposit rates when rates increase and quicker to cut them when rates fall. This asymmetry implies that the deposits channel may be stronger (in the sense of larger spread widening) during tightening than during easing, creating non-linear aggregate effects of monetary policy on liquidity supply. Formally testing this asymmetry using the DSS framework — and quantifying its implications for the liquidity premium's cyclical behaviour — would be a valuable extension.

Extension 5: International Cross-Country Evidence

The deposits channel is tested exclusively on U.S. data, where banking is predominantly private, deposit markets are geographically segmented, and depositor protection is moderate. A cross-country analysis would test whether the channel's strength is a general feature of market-based banking or is specific to the U.S. institutional context.

Thought experiment. Consider two stylised banking systems:

- **System A (US-type):** Privately owned banks, competitive deposit markets, moderate concentration, limited state guarantees beyond FDIC.
- **System B (State-bank dominated):** State-owned banks, administered deposit rates, implicit full deposit guarantees, no profit motive in deposit-taking.

In System B, the deposits channel should be muted or absent — state banks do not maximise deposit rents, so the rate-market-power nexus that drives DSS's mechanism does not apply. A panel of countries with varying degrees of state bank ownership, deposit guarantee generosity, and market concentration — using BIS banking statistics, ECB supervisory data, and national central bank rate data — could provide clean cross-sectional identification of the institutional preconditions for the deposits channel.

8. Contemporary Relevance and Policy Implications

8.1 The 2022–2023 Federal Reserve Tightening Cycle

The Federal Reserve’s 2022–2023 tightening cycle — raising the Fed funds rate from near-zero to over 5% in approximately 18 months, the fastest pace since 1981 — provides a rich out-of-sample test of DSS’s framework. Table 4 compares DSS’s theoretical predictions to observed outcomes.

Table 4: DSS Predictions vs. Observed Outcomes: 2022–2023 Tightening Cycle

DSS Prediction	Observed (2022–23)	Assessment
Large deposit outflows	~\$1 trillion decline in U.S. bank deposits	Confirmed
Substitution into money markets	MMF assets rose ~\$1.2 trillion	Confirmed
Deposit spread widening	Observed, heterogeneous across bank types	Confirmed
Lending contraction	New loan growth slowed substantially from 2022 peaks	Broadly confirmed
Speed of outflow adjustment	Faster than 1994–2013 historical patterns	Faster than predicted
Uniform response by bank type	Large disparities between large/small banks and by depositor sophistication	More heterogeneous than model

The directional predictions of the deposits channel were broadly confirmed. However, the speed and concentration of outflows was faster and more heterogeneous than the 1994–2013 model parameters would predict. Digital banking had structurally reduced switching frictions: depositors were more attentive and more capable of rapid reallocation than the model’s representative agent assumes. Large banks retained deposits more effectively than smaller institutions, partly reflecting the implicit “too big to fail” safety premium, and partly reflecting their broader product offerings that substituted within the banking relationship.

8.2 Silicon Valley Bank: The Deposits Channel in Extremis

The collapse of Silicon Valley Bank in March 2023 represents a natural stress experiment for DSS’s theoretical framework — and reveals its most important limitation.

SVB's depositor base exhibited the polar opposite of the characteristics that generate market power in the DSS model.

Table 5: SVB vs. the DSS High-Market-Power Depositor Profile

Characteristic	DSS High Market Power	SVB Reality
Depositor sophistication	Low; retail; inattentive	Extreme; VCs, tech CFOs; highly informed
Switching costs	High; geographic, behavioural	Near-zero; digital; one-day transfers
Network effects	None; depositors are diffuse	Dense; VC community shares information
Insurance coverage	Mostly covered by FDIC	>90% above FDIC \$250K limit
Rate sensitivity	Low; slow to respond	Extremely high; continuous monitoring
Deposit concentration	Dispersed across many depositors	Concentrated; top accounts = large %

When the Fed raised rates and SVB's mark-to-market losses on its long-duration bond portfolio became publicly known in early March 2023, depositor demand elasticity was essentially infinite. A classic bank run unfolded in hours: an estimated \$42 billion in deposits was withdrawn in a single day, with digital infrastructure eliminating the queuing frictions that historically slowed bank runs.

The SVB episode is not a refutation of the deposits channel — it is its limiting case with inverted parameters. DSS model the gradual, aggregate response of a representative depositor with meaningful switching costs. SVB illustrates what happens at the opposite extreme: zero switching costs, maximal sophistication, and concentrated information networks. The implication is that the deposits channel framework, as currently specified, cannot simultaneously model the slow-moving aggregate dynamics that DSS document and the discontinuous run dynamics that SVB exhibited. A unified framework incorporating heterogeneous depositors and strategic complementarities in withdrawal decisions (in the tradition of [Diamond and Dybvig 1983](#)) would be a significant theoretical contribution.

8.3 Policy Implications

8.3.1 Implications for Central Banks

The deposits channel implies that monetary tightening is more powerful than New Keynesian models suggest: it operates through a supply-side contraction in liquidity

in addition to the standard demand-side suppression via interest rates. The level of the short rate — not merely its deviation from the neutral rate — directly affects deposit supply, lending, and the liquidity premium. This provides theoretical grounding for two empirically observed but previously theoretically unexplained central bank behaviours: the practice of gradual rate adjustment (Bernanke, 2004; Stein and Sunderam, 2015), and the observed sensitivity of financial markets to rate level changes even when fully anticipated.

In a world of declining bank market power due to fintech disruption and potential CBDC introduction, the deposits channel may weaken structurally, reducing the potency of rate-based monetary transmission and increasing the relative importance of quantitative balance sheet tools. This has important implications for the design of exit strategies from near-zero rate environments and for the calibration of rate paths required to achieve given macroeconomic targets.

8.3.2 Implications for Macroprudential Regulation

Deposit spread betas should be incorporated into bank stress testing frameworks as a measure of rate sensitivity that complements traditional asset duration analysis. A bank with high spread beta will experience larger funding shocks under a given rate scenario, and this risk is not captured by standard duration-matching or interest rate risk metrics focused on the asset side.

Furthermore, bank mergers that increase deposit market concentration have a dual effect: they strengthen the deposit franchise and improve margins in normal times, but simultaneously increase the magnitude of funding shocks during monetary tightening cycles. Standard antitrust analysis that focuses on loan pricing effects of bank mergers may systematically understate the macroprudential costs of increased deposit market concentration.

The 90% correlation between deposit spreads and the T-bill liquidity premium suggests that monitoring deposit market conditions can serve as an early-warning indicator of broader liquidity stress in financial markets, ahead of observable distress in credit or repo markets. Regulators could usefully incorporate aggregate deposit spread measures into financial stability monitoring frameworks.

8.3.3 Implications for CBDC Design

The deposits channel analysis generates sharp predictions for the consequences of a retail CBDC. A CBDC paying a policy-linked rate would serve as a perfect outside option for depositors, eliminating geographic and behavioural switching frictions instantaneously. In the short run, this would dramatically accelerate and intensify deposit outflows during rate hiking cycles. In the long run, it would compress bank

market power toward zero, potentially eliminating the deposit spread beta and with it the deposits channel as a transmission mechanism. Policymakers designing CBDC systems must weigh these effects against the efficiency gains from improved monetary transmission. The optimal design likely involves a CBDC that is restricted in the quantity households can hold or pays a below-market rate, preserving some role for bank deposit markets while improving the transmission of monetary policy at the margin.

9. Conclusion

Drechsler, Savov and Schnabl (2017) make a landmark contribution to monetary economics by providing a micro-founded, empirically validated mechanism for the bank lending channel that does not rely on the implausible reserve requirements that had undermined prior theories for decades. Their central insight — that banks' market power over local deposit markets is endogenous to the level of the Fed funds rate, allowing them to capture rents from depositors during rate hiking cycles — is both theoretically elegant and empirically powerful.

The paper's primary contributions are threefold. First, it provides a clean identification strategy that isolates the deposit supply effect from confounding changes in lending opportunities, using within-bank variation across branches with different levels of local market concentration and validated by an event study with precise timing. Second, it establishes quantitative magnitudes sufficient to account for the entire transmission of monetary policy through bank balance sheets without reserve requirements. Third, it connects monetary policy to the broader liquidity premium in financial markets, extending the mechanism's reach well beyond the banking sector to all financial institutions that rely on liquid asset buffers.

The paper's primary limitations are its pre-fintech sample period, its representative household assumption, and the exogeneity of the wholesale funding cost parameter. These limitations are becoming more consequential as digital banking erodes geographic switching frictions, as the 2022–2023 tightening cycle demonstrated, and as the prospect of CBDCs raises fundamental questions about the future of bank market power in deposit markets.

Looking forward, the most important open questions are: How has the deposits channel evolved as switching costs have declined? Can the gradual aggregate dynamics the channel describes be reconciled with the discontinuous run dynamics observed at SVB in a unified theoretical framework? How would a retail CBDC alter the magnitude and speed of the channel? And does the channel operate symmetrically in easing and tightening episodes, or does the downward rigidity of

deposit rates create an inherent asymmetry?

The deposits channel remains one of the most compelling frameworks in modern monetary economics. The friction that powers it — depositor inertia and bank market power — is real, historically large, and consequential for both the real economy and financial markets. Understanding its evolution in the digital-banking era is among the most important tasks at the frontier of monetary economics and banking research.

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